

JEE MAIN REVISION MASTER: ELASTICITY 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Elasticity - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Elasticity - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Kcet Elasticity.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "There are two wires of same material and same length while the diameter of second wire is two times the diameter of the first wire. Then the..." [Ans: D]
- [2025 Shift] Q#2 (MCQ): "Two wires A and B are made of same material. Their diameters are in the ratio of 1:2 and lengths are in the ratio of 1:3 . If they are s..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "A thick metal wire of density ρ and length L is hung from a rigid support. The increase in length of the wire due to its own weight i..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

There are two wires of same material and same length while the diameter of second wire is two times the diameter of the first wire. Then the ratio of extensions produced in the wires by applying same load will be

- (A) 1:1
- (B) 1:2
- (C) 2:1
- (D) 4:1 **[Correct]**

Correct Answer: D

Solution Analysis: For a wire, the extension produced by a load is given by $\Delta L = (FL / AY)$ Here, F is the load applied, L is the length of the wire, A is the area of cross-section, Y is Young's modulus. Both wires have the same material, so Y is the same. Both have the same length, and the same load is applied. So, $\Delta L \propto (1 / A)$ Now, area of cross-section of a wire is $A = \frac{\pi d^2}{4}$ So area is proportional to the square of diameter: $A \propto d^2$ Given, $d_2 = 2d_1$ So, $A_2 = 4A_1$ Since extension is inversely proportional to area, $\frac{\Delta L_1}{\Delta L_2} = \frac{A_2}{A_1}$